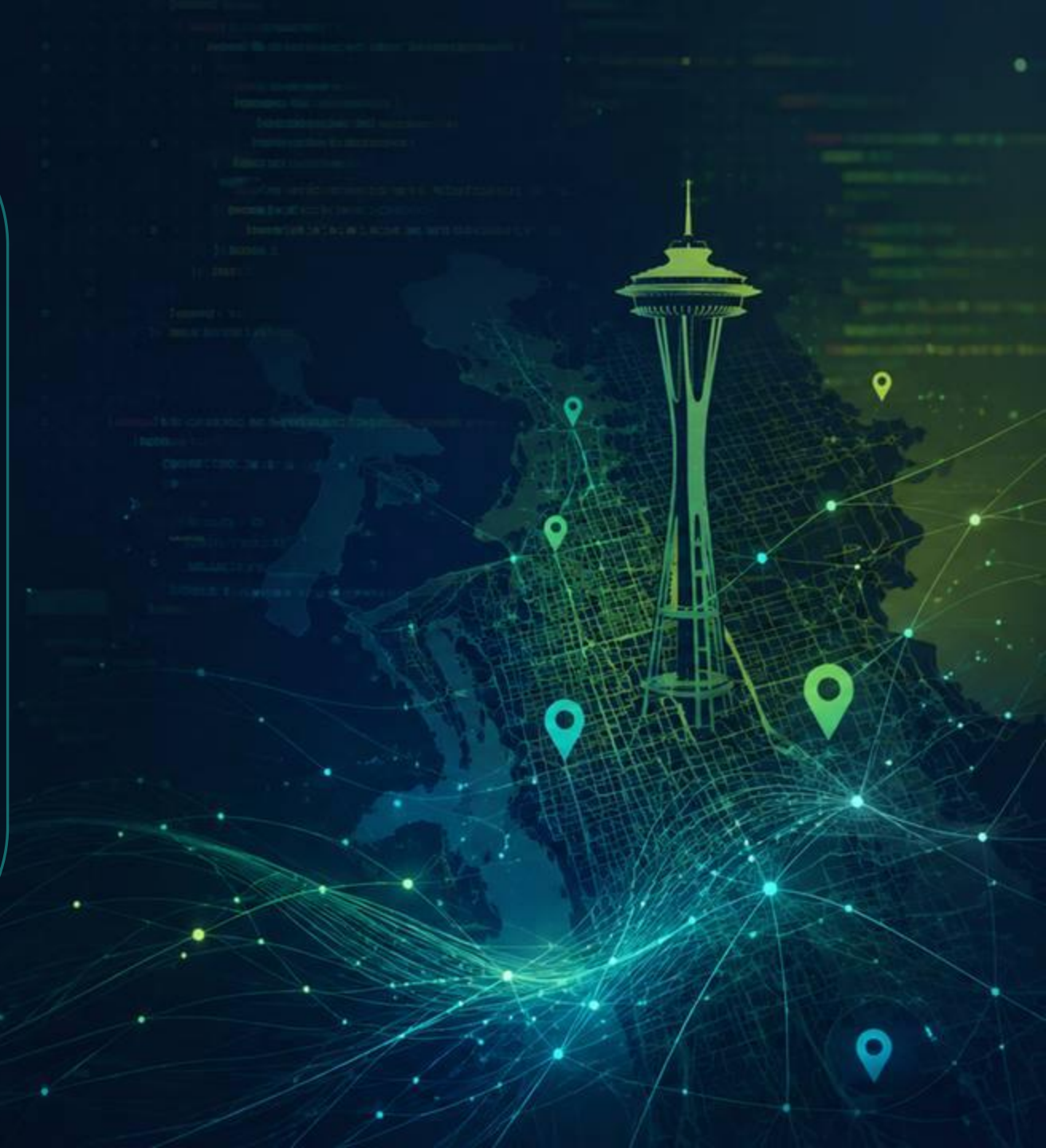


WEB WORKERS

Keep the Main Thread Free with Web Workers

Courtney Yatteau · Esri · CascadiaJS 2026



THE PLAN

Today's path

Feel the freeze.
Prove it.
Move the work.

1 Feel the freeze

Make the stutter obvious

2 Prove the blocker

Code + DevTools receipts

3 Move the work

Main thread $\Rightarrow$ worker

4 Add guardrails

Progress · cancel · debounce

5 Map it

Seattle ArcGIS proof

WHY IT MATTERS

Jank is a workload problem.

Users feel it before we diagnose it.

Delayed clicks

Frozen motion

Laggy interactions

DevTools sees: one big JS task

THREAD

A practical rescue path.



LIVE DEMO 01

THREAD: T + H

DEMO

01

Trigger the stutter

Just the browser under pressure.

WATCH FOR

Heartbeat Freezes

Click waits and typing box

One front desk.

The main thread handles the experience.

Clicks

Paint

UI
updates

Heavy JS
blocks the line

Not a second UI.

A separate workspace for data chores.

Main thread

DOM + UI
clicks + paint

request →

← result

Worker thread

plain data
calculation

LIVE DEMO 02

THREAD: R

DEMO

02

Move the work

Same workload. Different thread.

MAIN

freezes

WORKER

breathes

Make messages boring.

Boring is good when two threads talk.

```
{  
  type: "ANALYZE",  
  requestId: "req-001",  
  payload: {...}  
}
```

PROGRESS

DONE

CANCELLED

ERROR

LIVE DEMO 03

THREAD: E + A

DEMO

03

Progress + cancel

A worker job with visible messages.

message log

Start

Cancel

Restart

request IDs keep
order

GUARDRAILS

Workers help. Design still matters.

A worker is only one part of the pattern.

Debounce

Wait

Progress

Inform

Cancel

Stop stale

LIVE DEMO 04

THREAD: A

DEMO

04

Fast filters

A worker filter flow with debounce.

search: parks...

Typing waits

Old results ignored

What are we scoring?

Park boundary records for review priority

Question

Which visible boundary records should be reviewed first?

Size

larger footprint

bigger spatial impact

Complexity

more rings + vertices

harder to inspect

Data confidence

lower confidence

Review more carefully

Visible boundaries → score + sort records → highlight top results

DECIDE

Keep the experience on the main thread

Move expensive data chores when they earn the trip.

Good fit

Plain data in, plain data out
UI needs to stay alive

Probably overkill

Tiny calculations
DOM work
Normal async calls

Trigger · Hunt · Relocate · Establish · Add · Decide

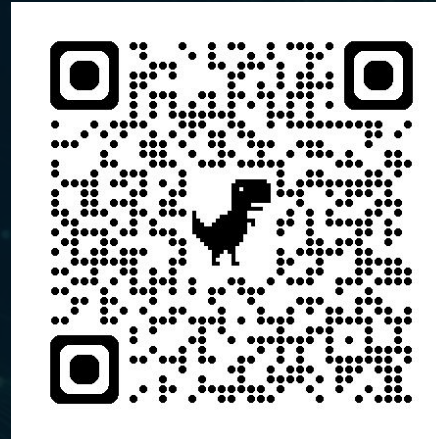
Thank you, CascadiaJS!

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<https://github.com/cyatteau/cascadiajs-2026-web-workers>